

# class practice

Subject: Physics | Chapter: electrostatics | Difficulty: medium

**Question 1:** Two point charges,  $q_1 = +2 \mu\text{C}$  and  $q_2 = -4 \mu\text{C}$ , are separated by a distance of 10 cm. What is the force between them?

**Options:**

- A) 0.72 N (attractive)
- B) 7.2 N (attractive)
- C) 0.72 N (repulsive)
- D) 7.2 N (repulsive)

**Question 2:** A uniformly charged sphere of radius  $R$  carries a total charge  $Q$ . What is the electric field at a distance  $r > R$  from the center of the sphere?

**Options:**

- A)  $kQ/r$
- B)  $kQ/r^2$
- C)  $kQ/R$
- D)  $kQ/R^2$

**Question 3:** What is the electric potential at a distance of 5 cm from a point charge of  $10 \mu\text{C}$ ?

**Options:**

- A)  $1.8 \times 10^6 \text{ V}$
- B)  $1.8 \times 10^5 \text{ V}$
- C)  $3.6 \times 10^6 \text{ V}$
- D)  $3.6 \times 10^5 \text{ V}$

**Question 4:** An electron is released from rest in a uniform electric field. Which of the following statements is TRUE?

**Options:**

- A) It will experience no force.
- B) It will move with constant velocity.
- C) It will accelerate uniformly.
- D) It will move in a circular path.

**Question 5:** What is the electric flux through a closed surface that encloses a net charge of  $10\ \mu\text{C}$ ?

**Options:**

- A)  $1.13 \times 10^6\ \text{Nm}^2/\text{C}$
- B)  $1.13 \times 10^5\ \text{Nm}^2/\text{C}$
- C)  $8.85 \times 10^{-8}\ \text{Nm}^2/\text{C}$
- D)  $8.85 \times 10^{-7}\ \text{Nm}^2/\text{C}$

**Question 6:** A capacitor has a capacitance of  $2\ \mu\text{F}$  and is charged to a potential difference of  $100\ \text{V}$ . What is the energy stored in the capacitor?

**Options:**

- A)  $0.01\ \text{J}$
- B)  $0.1\ \text{J}$
- C)  $1\ \text{J}$
- D)  $10\ \text{J}$

**Question 7:** What is the effect on the capacitance of a parallel plate capacitor if the distance between the plates is doubled?

**Options:**

- A) It doubles.
- B) It halves.
- C) It remains unchanged.
- D) It quadruples.

**Question 8:** Which of the following is NOT a unit of electric field strength?

**Options:**

- A)  $\text{N/C}$
- B)  $\text{V/m}$
- C)  $\text{J/C}$
- D)  $\text{C/m}$

**Question 9:** A dielectric material is inserted between the plates of a charged capacitor. What happens to the capacitance?

**Options:**

- A) It decreases.
- B) It increases.
- C) It remains unchanged.
- D) It becomes zero.

**Question 10:** Lightning is an example of which electrostatic phenomenon?

**Options:**

- A) Electrostatic induction
- B) Electrostatic discharge
- C) Charging by conduction
- D) Charging by friction

**Question 11:** Two identical conducting spheres carry charges of  $+5\ \mu\text{C}$  and  $-3\ \mu\text{C}$ . They are brought into contact and then separated. What is the charge on each sphere?

**Options:**

- A)  $+1\ \mu\text{C}$
- B)  $+2\ \mu\text{C}$
- C)  $+4\ \mu\text{C}$
- D)  $+5\ \mu\text{C}$

**Question 12:** A charge of  $4\ \mu\text{C}$  is placed at the center of a cube. What is the electric flux through one face of the cube?

**Options:**

- A)  $4.52 \times 10^4\ \text{Nm}^2/\text{C}$
- B)  $1.81 \times 10^5\ \text{Nm}^2/\text{C}$
- C)  $7.24 \times 10^5\ \text{Nm}^2/\text{C}$
- D)  $1.44 \times 10^5\ \text{Nm}^2/\text{C}$

**Question 13:** What is the potential energy of two point charges,  $q_1 = 2 \mu\text{C}$  and  $q_2 = -3 \mu\text{C}$ , separated by a distance of 5 cm?

**Options:**

- A) -1.08 J
- B) +1.08 J
- C) -0.108 J
- D) +0.108 J

**Question 14:** Xerography is based on which electrostatic principle?

**Options:**

- A) Electrostatic induction
- B) Electrostatic discharge
- C) Charging by conduction
- D) Charging by friction

**Question 15:** A parallel plate capacitor has a dielectric constant of 3. If the dielectric is removed, what happens to the capacitance?

**Options:**

- A) It increases by a factor of 3
- B) It decreases by a factor of 3
- C) It remains the same
- D) It doubles

**Question 16:** What is the electric field inside a conductor in electrostatic equilibrium?

**Options:**

- A) Zero
- B) Constant
- C) Variable
- D) Infinite

**Question 17:** Two point charges of equal magnitude and opposite sign are placed close together. This system is called a:

**Options:**

- A) Monopole
- B) Dipole
- C) Tripole
- D) Quadrupole

**Question 18:** The process of transferring charge from one object to another without direct contact is called:

**Options:**

- A) Conduction
- B) Induction
- C) Friction
- D) Convection

**Question 19:** A Van de Graaff generator is used to:

**Options:**

- A) Measure electric potential
- B) Generate high voltage
- C) Measure electric current
- D) Store electric energy

**Question 20:** If the potential difference across a capacitor is doubled, the energy stored in the capacitor:

**Options:**

- A) Is halved
- B) Remains the same
- C) Is doubled
- D) Is quadrupled